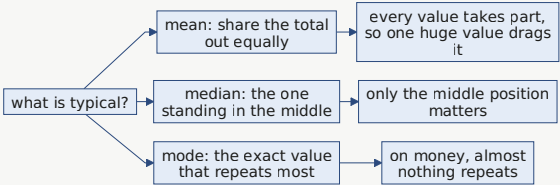


typical, spread and denominators

Week 1, Day 4. Same eight panels with about a quarter of the cells removed.

1 the three "typical" numbers



	Answers	Reach for it when
Mean	_____	The shape is roughly even
Median	The record in the middle of the queue	_____
Mode	The value that repeats most	_____

Crux. nobody is right, they answer _____, and the asker rarely says which one they meant.

2 the one-line skew test

_____ / _____		
Ratio	Shape	Send
Close to 1	_____	Either
Well above 1	Right tail, something large	_____
Well below 1	_____	Median
Run it first on any column you have never seen.		

3 what one record does

seven orders	mean Rs 1,676.43	median Rs 1,310
swap the top for 480000	mean Rs _____	median Rs _____
Crux. the mean _____, the median moved by _____. Not "a little".		

4 the fence

q1, _, q3 = statistics.quantiles(values, n=4)
iqr = _____
upper = _____
lower = _____
Crux. a fence is a _____, never a _____. An outlier is a finding to investigate before it is a row to delete.

5 grouping without knowing the categories

```
buckets = {}
for r in orders:
    key = r[field]
    if _____:
        buckets[key] = {"count": 0, "amounts": [], "returned": 0}
    buckets[key]["count"] += 1
    buckets[key]["amounts"]._____ (r["amount"])
    if r["status"] == "_____":
        buckets[key]["returned"] += 1
```

Short form of the same idea: counts[key] = counts._____ (key, _____) + 1

Crux. every hard-coded list of categories is _____. Hard-code three and Kalpa has four, and you get _____: '_____'.

6 count first, then median

Accumulate one record at a time	Needs the whole group first
count, sum, returned count, _____,	_____, _____, the fence

Crux. there is no _____.	

7 the honest sentence

[_____]	[_____]	[_____]
Write the sentence for Business, which had 1 returned of 9 orders:		
Crux. the third part is the one that goes missing between your notebook and somebody else's slide.		

8 before you send a rate

Check	If it fails
_____	It is not ready
Any group under your stated floor	_____
Ranked groups of very different sizes	Say what the ranking is partly measuring
A mean quoted on a money column	_____
A difference being called real	_____
THE THREE LINES WORTH MEMORISING	
1. The mean was _____ and the description was _____.	
2. On a money column, send the _____, and say that is what you sent.	
3. Every rate carries its _____, or it lies for you while you are not in the room.	

THE WORDS ON THIS SHEET

Mean The total shared out equally, which every value takes part in

Median The value standing in the middle of the sorted column

Mode The exact value that repeats most, which on money repeats twice in 44

Range Maximum minus minimum, built from the two least typical values

Interquartile range The spread of the middle half, and the basis of the fence

Skew Which side of the median the tail is on

Denominator The count a rate rests on, which travels with it

Trust floor A record count below which you decline to rank, chosen and defended